

SHAP, Causal SHAP, and Causal Responsibility Attribution

Summary of Key Results and Arguments

1. The Running Model

All results use the following linear Gaussian chain:

$$X \sim \mathcal{N}(0.5, 0.25)$$

$$M = X + \epsilon_M, \quad \epsilon_M \sim \mathcal{N}(0, 0.1)$$

$$Y = M + \epsilon_Y, \quad \epsilon_Y \sim \mathcal{N}(0, 0.1)$$

Causal graph: $X \rightarrow M \rightarrow Y$. All variables jointly Gaussian. All conditional expectations are exact.

Key distributional facts:

- $\text{Var}(X) = 0.25, \text{Var}(M) = 0.35, \text{Var}(Y) = 0.45$
- $\text{Cov}(X, M) = 0.25, \text{Cov}(X, Y) = 0.25, \text{Cov}(M, Y) = 0.35$

Prediction Functions

For each target variable, the optimal predictor is the conditional expectation:

Target Y, features {X, M}:

$$f_Y(X, M) = E[Y|X, M] = M$$

X drops out because M screens X from Y (d-separation).

Target M, features {X, Y}:

$$f_M(X, Y) = E[M|X, Y] = 0.5X + 0.5Y$$

M is exactly halfway between X and Y because noise variances are equal at each step.

Target X, features {M, Y}:

$$f_X(M, Y) = E[X|M, Y] = E[X|M] = 0.5 + \frac{5}{7}(M - 0.5)$$

Y drops out exactly because $X \perp Y \mid M$ (d-separation: M blocks the only path).

The coefficient $5/7 = \text{Cov}(X, M) / \text{Var}(M) = 0.25 / 0.35$ reflects that M is a noisy signal of X.

2. Shapley Values: Definition

For a cooperative game (N, v) the Shapley value for player i is:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N|-|S|-1)!}{|N|!} [v(S \cup \{i\}) - v(S)]$$

The weight counts: what fraction of all orderings of N have exactly the members of S appearing before i ?

For $|N|=2$ this simplifies to: each ordering gets weight $1/2$.

SHAP value function:

$$v(S) = E_{\{X_{\bar{S}}\}}[f(X_S = x_S, X_{\bar{S}})]$$

Missing features are marginalized over their distribution. SHAP decomposes:

$$f(x) = f_0 + \sum_i \phi_i$$

where $f_0 = E[f(X)]$.

Key point: SHAP explains $f(x)$ — the model prediction — not the actual outcome Y . This is forced by the local accuracy axiom in Lundberg & Lee (2017). The residual $Y - f(x)$ (the noise) is invisible to SHAP by construction.

3. Plain SHAP Results

Instance: $X=1$, $M=1$, $Y=1$. Baseline = 0.5 throughout.

Target Y , features $\{X, M\}$

Coalition	Value	Reason
$\{\}$	0.5	$E[M]$
$\{X\}$	1	$E[M X=1] = 1$
$\{M\}$	1	$M=1$ directly
$\{X, M\}$	1	$f_Y(1,1) = 1$

$$\phi_X = 0.25, \quad \phi_M = 0.25$$

Target M, features {X, Y}

Coalition	Value	Reason
{}	0.5	$E[M]$
{X}	1	$E[Y X=1]=1$, so $0.5+0.5=1$
{Y}	0.889	$E[X Y=1]=0.778$, so $0.5 \times 0.778 + 0.5 = 0.889$
{X,Y}	1	$f_M(1,1)=1$

$$\phi_X = 0.306, \quad \phi_Y = 0.194$$

Target X, features {M, Y}

Coalition	Value	Reason
{}	0.5	$E[X]$
{M}	0.857	$0.5 + 5/7 \times 0.5$
{Y}	0.778	$E[X Y=1] = 0.5 + 0.25/0.45 \times 0.5$
{M,Y}	0.857	$X \perp Y M$ so same as {M}

$$\phi_M = 0.219, \quad \phi_Y = 0.139$$

Key Failure: Causal Direction Invisible

For target X, plain SHAP gives $\phi_Y = 0.139 > 0$ even though Y does not cause X. Y is statistically informative about X through the chain $X \rightarrow M \rightarrow Y$, and SHAP cannot distinguish “Y is informative about X” from “Y causes X.”

4. The Mediator Dilution Problem

Expanding $X \rightarrow M \rightarrow Y$ to $X \rightarrow M1 \rightarrow M2 \rightarrow Y$ while keeping joint (X,Y) distribution identical:
 $\sigma^2_1 = \sigma^2_2 = 0.05 \quad \text{\text{(half the original noise each)}}$

Since $f_Y = M2$, SHAP splits credit equally:

Model	ϕ_X	ϕ_M or ϕ_{M2}
$X \rightarrow M \rightarrow Y$	0.25	0.25
$X \rightarrow M1 \rightarrow M2 \rightarrow Y$	0.167	0.167
$X \rightarrow M1 \rightarrow M2 \rightarrow M3 \rightarrow Y$	0.125	0.125

In general with k mediators: $\phi_X = 0.5/(k+1)$

The attribution to X shrinks purely as a function of modeling granularity, not causal facts.

Telephone game illustration: X is the original message, M1, M2, ... are successive intermediaries each adding small distortions, Y is the final received message. Splitting one intermediary into two — while keeping total distortion constant — reduces X’s apparent responsibility under SHAP.

5. Causal Shapley Values

Definition (Heskes et al. 2020):

$$v_{\{\text{causal}\}}(S) = E[f(X) \mid \text{do}(X_S = x_S)]$$

By do-calculus:

- Descendants of S outside S: marginalize over $P(\cdot \mid \text{do}(X_S = x_S))$
- Non-descendants of S outside S: marginalize over marginal $P(\cdot)$

Everything else (weights, formula, axioms) unchanged.

Causal SHAP Results

Target Y, features {X, M}:

Coalition {X}: M is descendant of X, marginalize over $p(M \mid \text{do}(X=1)) = N(1, 0.1)$:

$$v(\{X\}) = 1$$

No confounding so same as plain SHAP. Result: $\phi_X = \phi_M = 0.25$.

Target M, features {X, Y}:

Coalition {X}: Y is descendant of X, marginalize over $p(Y \mid \text{do}(X=1)) = N(1, 0.2)$:

$$v(\{X\}) = 1$$

Coalition $\{Y\}$: X is NOT descendant of Y , marginalize over $p(X)$:

$$v(\{Y\}) = E_{X \sim p(X)}[0.5X + 0.5] = 0.75$$

This differs from plain SHAP (which gave 0.889). Result: $\phi_X = 0.375$, $\phi_Y = 0.125$.

Target X , features $\{M, Y\}$:

Coalition $\{Y\}$: $\text{do}(Y=1)$ cuts $M \rightarrow Y$, M is not affected. f_X doesn't depend on Y :

$$v(\{Y\}) = 0.5 = v(\{\})$$

Y adds nothing to any coalition. Result: $\phi_M = 0.357$, $\phi_Y = 0$.

Full Comparison Table

Target	Feature	Plain SHAP	Causal SHAP
Y	X	0.25	0.25
Y	M	0.25	0.25
M	X	0.306	0.375
M	Y	0.194	0.125
X	M	0.219	0.357
X	Y	0.139	0

Key improvement: Causal SHAP correctly zeros out ϕ_Y for target X — Y has no causal effect on X and $\text{do}(Y=y)$ does not affect M 's distribution. The causal direction $X \rightarrow M \rightarrow Y$ is clearly visible in the table.

Residual problem: $\phi_Y = 0.125$ persists for target M. Y appears in $f_M(X,Y) = 0.5X + 0.5Y$ as a statistical signal about M. Causal SHAP cannot remove this because it lives in the prediction function f , not in the marginalization.

6. Direct and Indirect Effect Decomposition

Heskes et al. decompose each $\phi_i(\pi)$ into:

Direct effect: change in prediction from fixing $X_i = x_i$ while holding distribution of other out-of-coalition features fixed under $\text{do}(X_S = x_S)$.

Indirect effect: additional change in distribution of out-of-coalition descendants caused by $\text{do}(X_i = x_i)$.

Note: “direct” and “indirect” here refer to information flow through f , not to direct edges in the causal DAG.

Decomposition Results

Target	Feature	Direct	Indirect	Total
Y	X	0	0.25	0.25
Y	M	0.25	0	0.25
M	X	0.25	0.125	0.375

M	Y	0.125	0	0.125
X	M	0.357	0	0.357
X	Y	0	0	0

Key observations:

1. **X's contribution to Y is entirely indirect (0.25 indirect, 0 direct):** correctly reflects that $X \rightarrow Y$ has no direct edge. X only reaches Y via M.
 2. **M's contribution to Y is entirely direct (0.25 direct, 0 indirect):** correctly reflects the direct edge $M \rightarrow Y$.
 3. **Y's contribution to M (0.125) is entirely direct and has no causal interpretation:** Y has no descendants so it can never have indirect effects. This attribution is purely epistemic — f_M uses Y as a statistical signal about M.
 4. **X's contribution to M has both direct (0.25) and indirect (0.125) components:** Direct because f_M depends on X explicitly. Indirect because setting $X=1$ shifts Y's distribution (Y is a descendant of X) which feeds into f_M through the $0.5Y$ term.
-

7. Residual Limitations for Causal Responsibility Attribution

7.1 SHAP Explains Predictions, Not Outcomes

At instance $X=0.5, M=0.5, Y=1$:

- $f_Y(0.5, 0.5) = 0.5 = \text{baseline}$
- Actual $Y = 1$
- $\phi_X = \phi_M = 0$

SHAP assigns zero responsibility to everything. The gap $Y - f(x) = 0.5$ is entirely due to ϵ_Y — pure noise. SHAP cannot see noise. Even though X and M affect the *probability* of $Y=1$ (different values of X would make $Y=1$ more or less likely), SHAP only explains the conditional expectation, not the realized value.

7.2 The Epistemic vs. Causal Direct Effect Problem

Causal SHAP eliminates spurious indirect effects in wrong causal directions. But it cannot eliminate spurious direct effects, because those come from f itself — which is derived from the observational joint distribution and encodes statistical dependencies regardless of causal direction.

7.3 No Responsibility Principle is Made Explicit

SHAP implements: *symmetric informational contribution, corrected for causal structure in the marginalization*. This does not correspond to standard philosophical responsibility principles:

- **Exogeneity:** assign responsibility to root causes (X), treat mediators as mechanisms
- **Proximate causation:** assign responsibility to most immediate cause
- **Path-specific:** decompose along causal paths, not features
- **Agency:** assign responsibility to variables under an agent's control
- **Actual causation:** responsibility tracks counterfactual dependence in the actual world

The equal split $\phi_X = \phi_M = 0.25$ for target Y is defensible under symmetric information but wrong under exogeneity and wrong under proximate causation. The choice of principle must be made external to the SHAP framework.

8. The Necessity/Sufficiency Model

Model Specification

Exogenous noise terms, all mutually independent:

$$U_X \sim \text{N}(0, 1)$$

$$U_Y \sim \text{N}(0, 1)$$

$$U_Z \sim \text{N}(0, 1)$$

$$U_O \sim \text{N}(0, 0.1)$$

Structural equations:

$$X = U_X$$

$$Z = U_Z$$

$$Y = Z + U_Y$$

$$O = 6XZ + 4Y + 4Z + U_O$$

Causal graph: $X \rightarrow O$ (via interaction), $Z \rightarrow O$ (direct and via interaction), $Z \rightarrow Y \rightarrow O$.

Instance $X = Y = Z = 1$ (consistent: $Z=1$, $U_Y=0$ gives $Y=1$)

$$O = 6 + 4 + 4 = 14$$

Necessity Analysis (counterfactual removal)

X: Set $X=0$, keep $Y=1$, $Z=1$:

$$O_{\{X=0\}} = 0 + 4 + 4 = 8. \quad \text{Drop: 6}$$

X's contribution lives entirely in $6XZ$. Removing X kills the interaction.

Y: Set $Y=0$, keep $X=1$, $Z=1$:

$$O_{\{Y=0\}} = 6 + 0 + 4 = 10. \quad \text{Drop: 4}$$

Y only affects $4Y$. Removing Y is a modest reduction.

Z: Set $Z=0$, keep $X=1$, $Y=1$:

$$O_{\{Z=0\}} = 0 + 4 + 0 = 4. \quad \text{Drop: 10}$$

Z appears in both $6XZ$ and $4Z$. Removing Z kills both terms.

Necessity ranking: $Z (10) > X (6) > Y (4)$

Sufficiency Analysis (standalone effect)

X alone ($Y=0$, $Z=0$):

$$O = 6 \cdot 1 \cdot 0 + 0 + 0 = 0$$

X has no direct path. X alone produces nothing.

Y alone ($X=0, Z=0$):

$$O = 0 + 4 \cdot 1 + 0 = 4$$

Y has a direct coefficient 4 independent of X and Z.

Z alone ($X=0, Y=0$):

$$O = 0 + 0 + 4 \cdot 1 = 4$$

Z has a direct coefficient 4. The XZ interaction vanishes without X.

Sufficiency ranking: $Y = Z (4) > X (0)$

Summary

Variable	Necessity (drop)	Sufficiency (standalone)	Interpretation
X	6	0	Necessary but not sufficient
Y	4	4	Sufficient but not necessary
Z	10	4	Both necessary and sufficient

X: Pure interaction variable. Needs Z to act. Its absence removes the dominant interaction term.

Y: Direct path variable. Works alone. Its absence only modestly affects O.

Z: Appears everywhere — direct path and interaction. Most necessary (largest drop) and equally sufficient as Y.

9. Key References

- Lundberg, S.M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *NeurIPS* 30.
- Heskes, T., Sijben, E., Bucur, I.G., and Claassen, T. (2020). Causal Shapley values. *NeurIPS* 33.
- Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*. Cambridge University Press.
- Shapley, L.S. (1953). A value for n-person games. *Contributions to the Theory of Games*, 2:307–317.